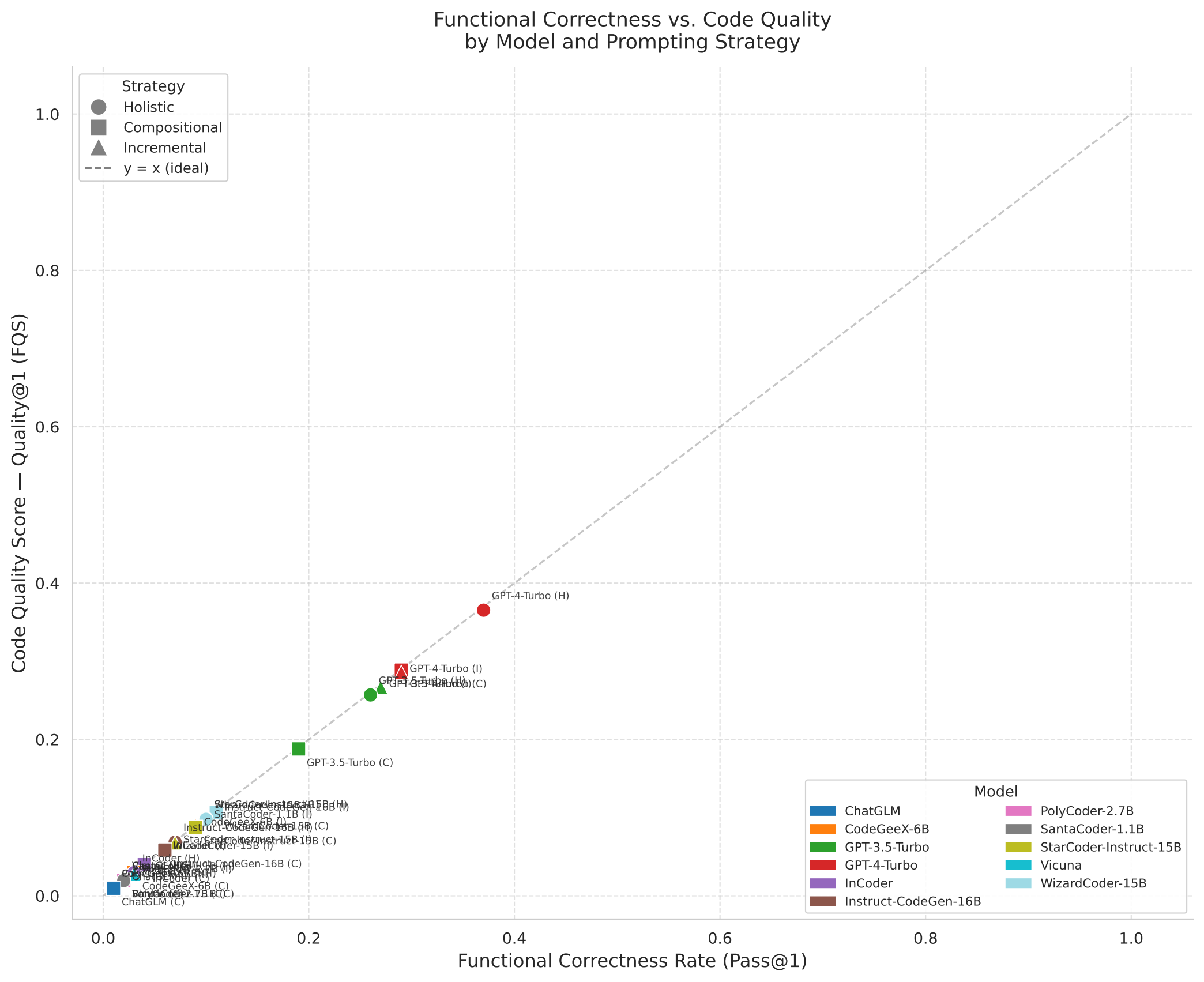
Functional Correctness vs. Code Quality
by Model and Prompting Strategy
Strategy
Holistic
Compositional
Incremental
y = x (ideal)
Model
ChatGLM
CodeGeeX-6B
GPT-3.5-Turbo
GPT-4-Turbo
InCoder
Instruct-CodeGen-16B
PolyCoder-2.7B
SantaCoder-1.1B
StarCoder-Instruct-15B
Vicuna
WizardCoder-15B
GPT-4-Turbo (H)
GPT-4-Turbo (I)
GPT-3.5-Turbo (C)
SantaCoder-1.1B (I)
CodeGeeX-6B (I)
InCoder (H)
CodeGeeX-6B (C)
ChatGLM (C)
Code Quality Score — Quality@1 (FQS)
Functional Correctness Rate (Pass@1)
0.0
0.2
0.4
0.6
0.8
1.0